



Sanctions Through Statistics

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PROJECT OVERVIEW

Is there a pattern behind Peps with sanctions?

- Sanctions are government bans on foreign companies, people and assets
- Politically Exposed Persons (PEP) data.



open sanctions

OUR APPROACH

BRAINSTORM

- Correlation between PEPS and sanctions
- Likelihood of being a potencial risk

$$TA = W_h * H + W_p * (\frac{C}{N})^\alpha$$

- How countries benefit from hidden data

OUR APPROACH

BRAINSTORM

- Correlation between PEPS and sanctions
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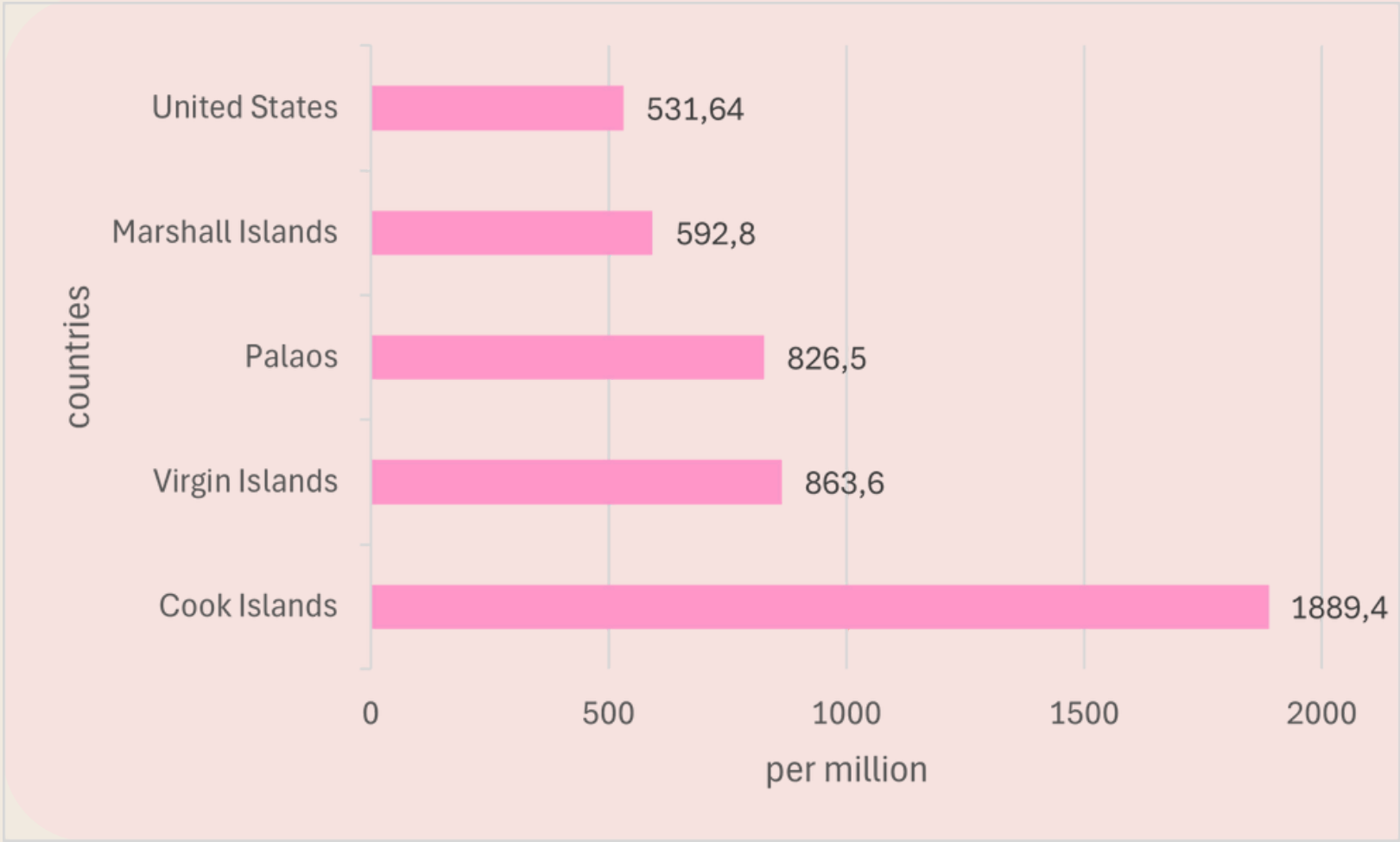
$$TA = W_h * H + W_p * (\frac{C}{N})^\alpha$$

- How countries benefit from hidden data

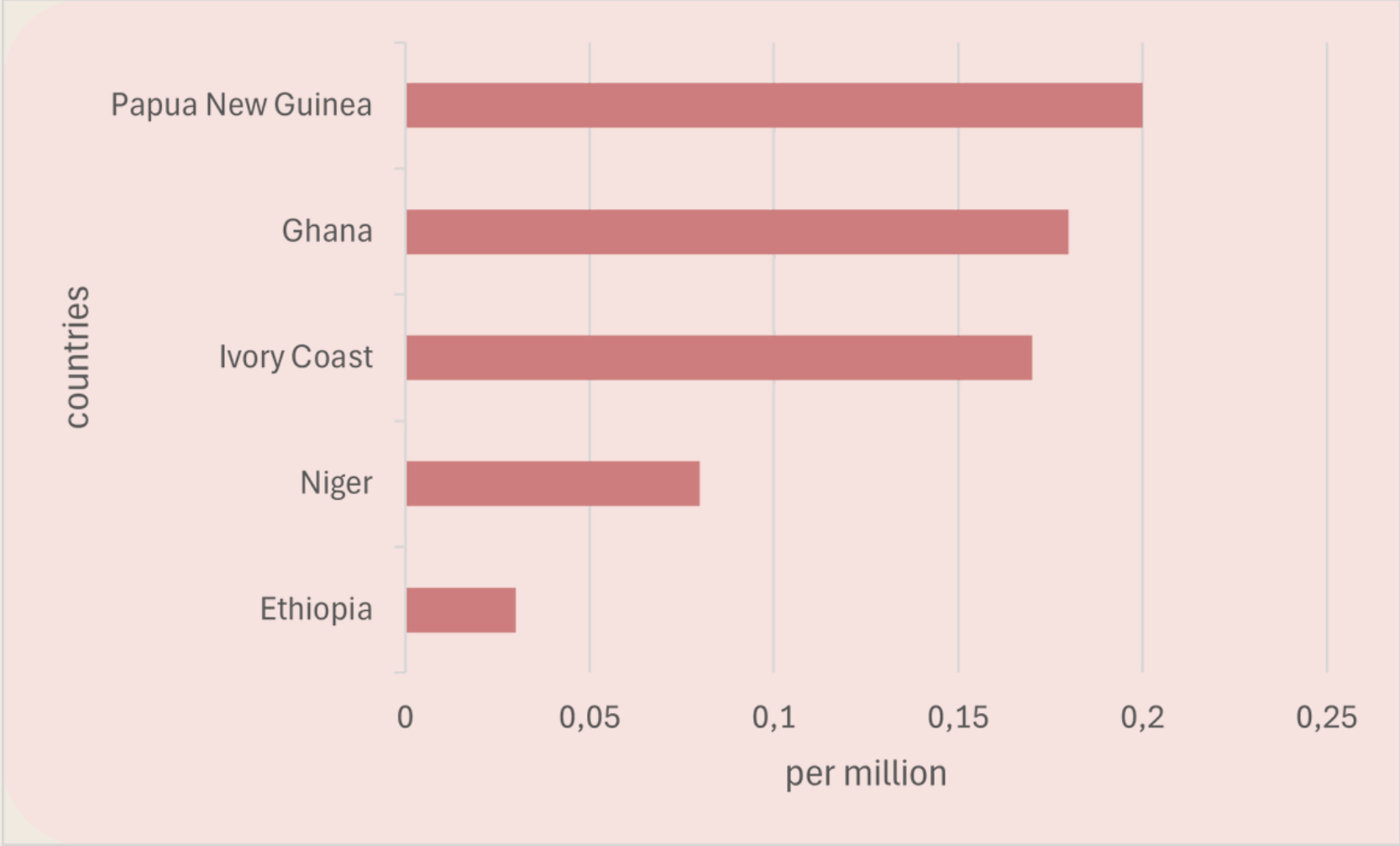
FINAL PROJECT

- Countries with higher sanctions coefficients based on population
- Full analytical workflow to show the relation between PEPS and sanctions

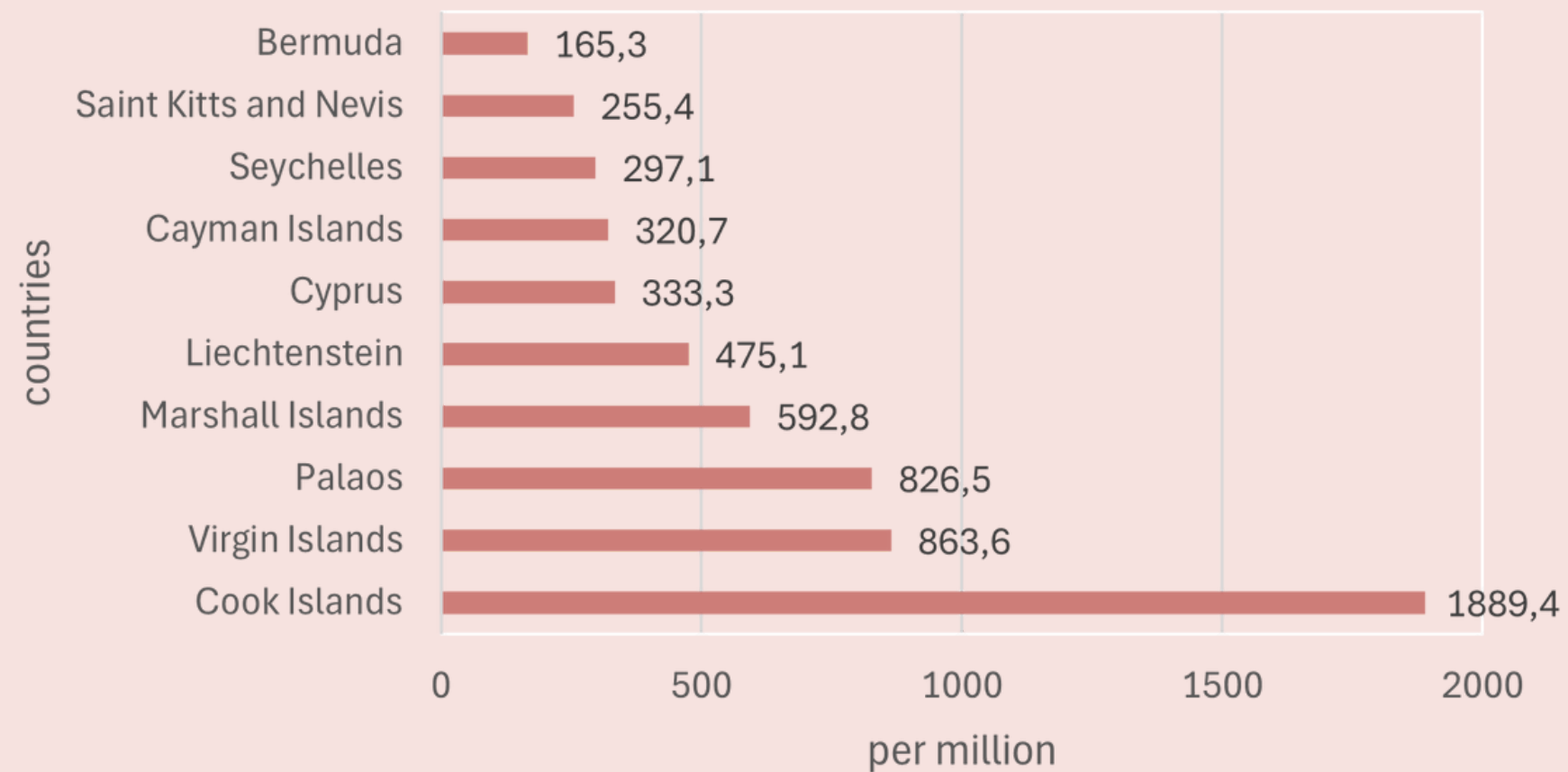
COUNTRIES WITH MOST SANCTIONS



COUNTRIES WITH LESS SANCTIONS



SMALLEST COUNTRIES WITH MOST SANCTIONS



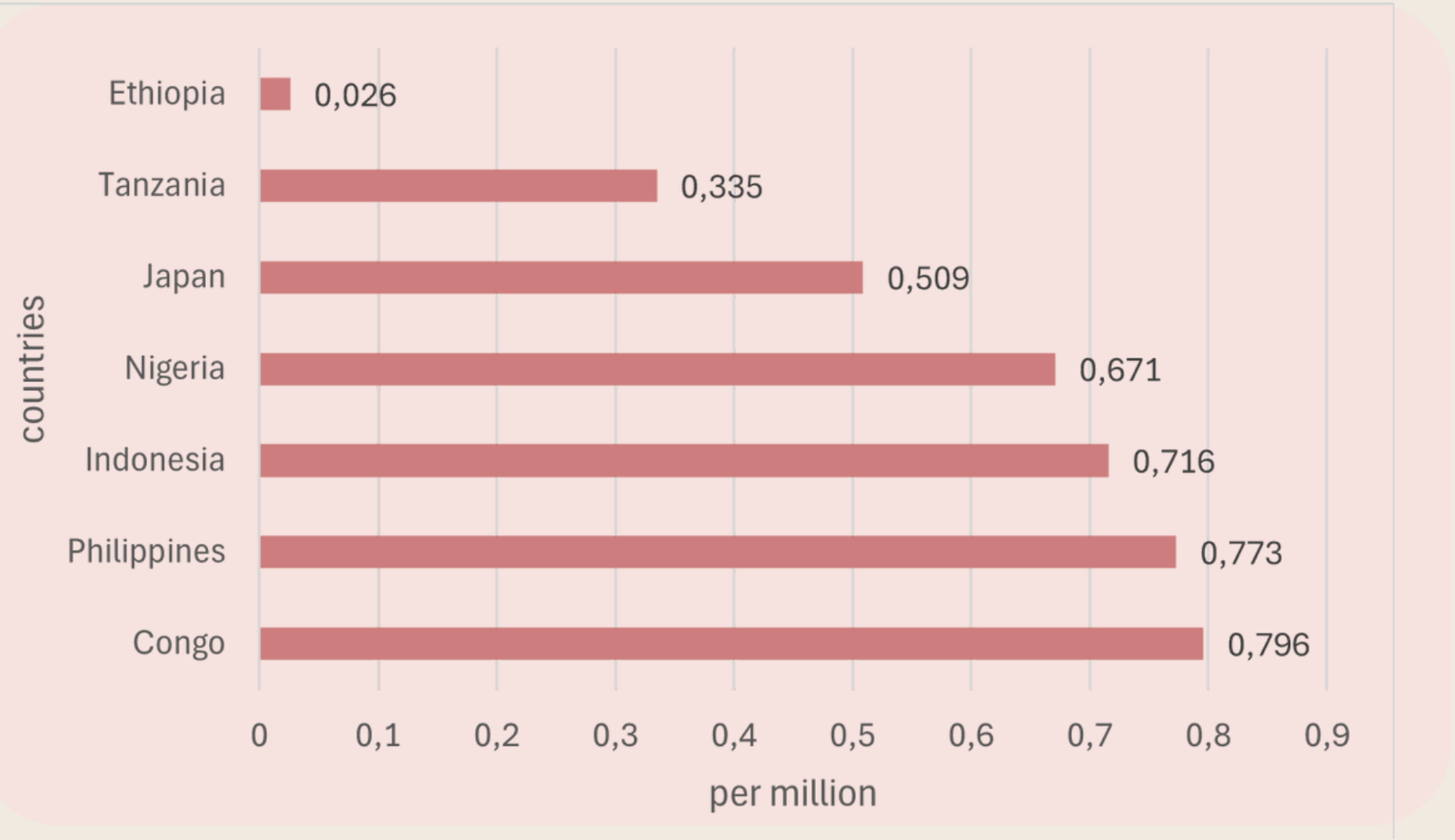
<1M: 46 countries, mean: 162.8/millon

1M-10M: 65 countries, mean: 24.9/millon

10M-50M: 63 countries, mean: 7.2/millon

50M+: 28 countries, mean: 34.6/millon

BIGGEST COUNTRIES WITH LOWEST SANCTIONS



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1M-10M: 65 countries, mean: 24.9/millon
10M-50M: 63 countries, mean: 7.2/millon
50M+: 28 countries, mean: 34.6/millon

DATA PROCESSING

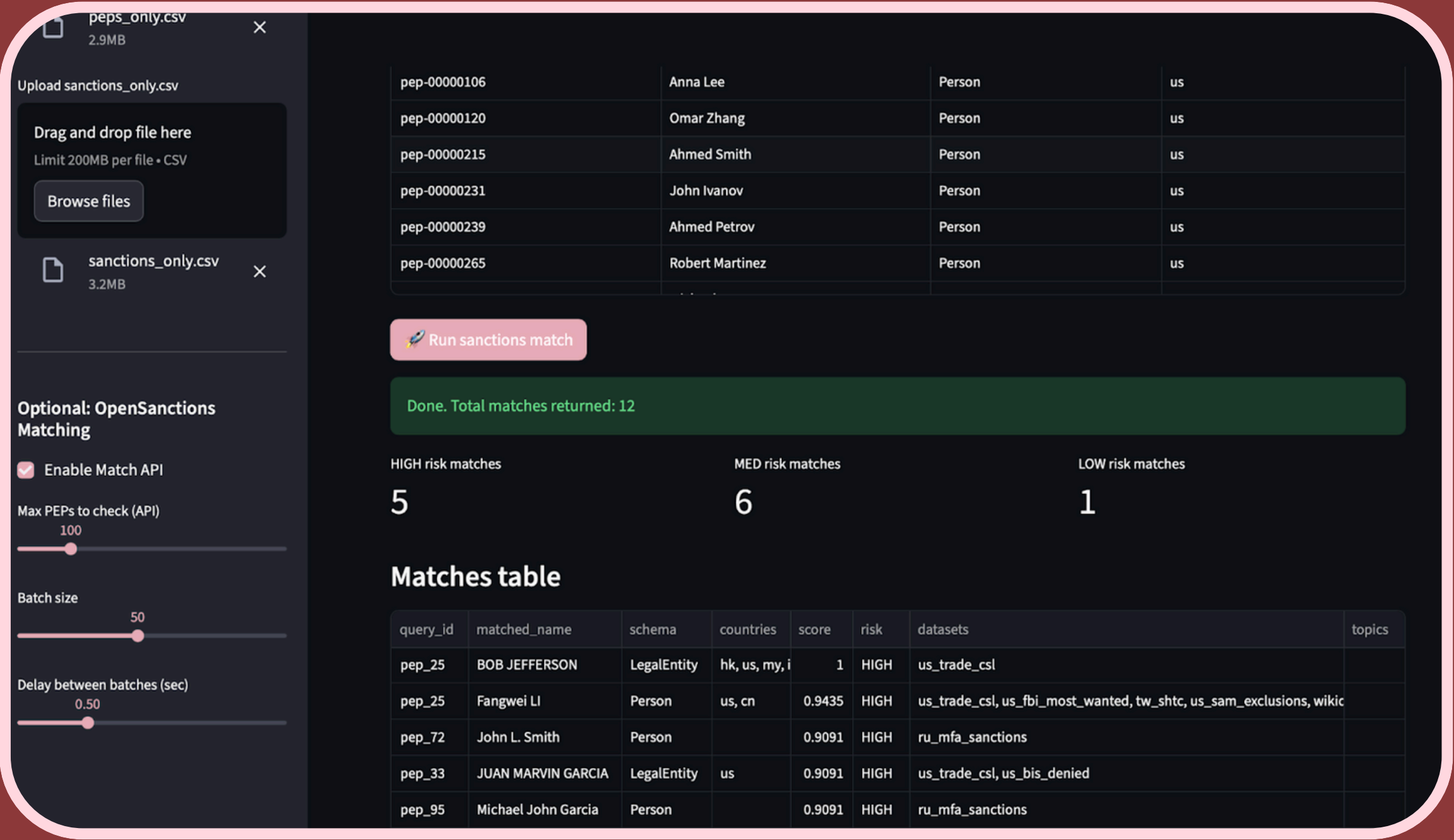
What we want: for the 10 most powerful countries, find how many PEPs have sanctions

- Download OpenSanctions' PEPs .csv file
- For X country, extract its PEPs
- Using an API, match the results with OpenSanctions' sanctions dataset
 - Filter to get accurate results
- Compute percentage

Relationship between PEPs and sanctioned entities across the ten most powerful countries

Country	PEPs	Matches	Percentage
United States US	58307	1828	3,14%
China CN	13276	882	6,64%
Russia RU	13950	3383	24,25%
United Kingdom GB	7038	70	5,48%
Germany DE	9248	25	0,37%
South Korea KR	2465	7	0,28%
France FR	59698	65	0,10%
Japan JP	2864	1	0,03%
South Arabia SA	543	33	6,08%
Israel IL	820	12	1,46%
Spain ES	1750	10	0,57%

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```
import os
import time
import pandas as pd
import streamlit as st
import altair as alt
import requests
```

```
st.set_page_config(
    page_title="PEPs & Sanctions Dashboard",
    page_icon="",
    layout="wide",
)

st.title(" PEPs & Sanctions Dashboard")
st.caption("Explore PEPs and sanctioned entities by country, and optionally run OpenSanctions matching.")
```

```
def norm_country_input(x: str) -> str:
    return (x or "").strip().upper()

def contains_country(series: pd.Series, code: str) -> pd.Series:
    """
    'countries' in your CSV can be:
    - "us"
    - "us,ca"
    - "[us,'ca']"
    We'll match by substring (robust to lists/comma-separated values).
    """
    code = (code or "").strip().lower()
    if not code:
        return pd.Series([False] * len(series), index=series.index)
    s = series.fillna("").astype(str).str.lower()
    return s.str.contains(code, na=False)
```



LIMITATIONS CONCLUSIONS AND FUTURE IMPROVEMENTS



DATA4GOOD